Solar Charging Station

Ved Sheth, Jack Bian, Sanjay Thornton

Overview

Functionality:

- Solar panel charges a LiPo battery to 3.7V
- Boost converter steps up voltage to 5V for a USB-C output port and a LDO(Low Dropout Regulator) steps it down to 3.3V to operate the ESP32 microcontroller
- USB-C output port is used to charge device
- ESP32 controls an output display that shows whether the connected device is charging and its battery percentage

Categories: Power, Actuation, Computation

BOM

Total cost per board: ~$82

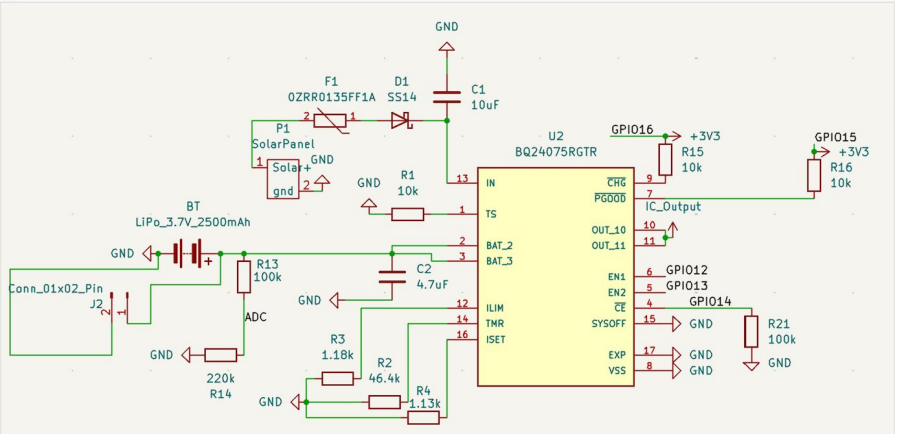
Qty (for 1 board)	Manufacturer Part #	Digi-key Part#/SKU	Mouser I	Price/	Purchase URL	Refere Desig	Description (include component value if applicable)
1	ESP32-S3-WROOM-1-N8	5407-ESP32-S3-WROOM-1		5.49	https://www.digi		WiFi 802.11a/b/g/n, Bluetooth v5.0 Transceiver Module 2.4GHz PCB Trace
1	BQ24075RGTR	296-38874-1-ND		2.36	https://www.digi		IC BATT CHG LI-ION 1CELL 16QFN
1	TPS61023DRLR	296-TPS61023DRLRCT-ND		1.08	https://www.digi		IC REG BOOST ADJ 3A SOT563
1	AP7333-33SRG-7	AP7333-33SRG-7DICT-ND		0.27	https://www.digi		IC REG LINEAR 3.3V 300MA SOT23R
1	SS14	SS14CT-ND		0.29	https://www.digi		DIODE SCHOTTKY 40V 1A DO214AC
1	0ZRR0135FF1A	5923-0ZRR0135FF1A-ND		0.45	https://www.digi		PTC RESET FUSE 60V 1.35A RADIAL
1	Product ID - 328	ADAFRUIT		14.95	Lithium Ion Polym		Lithium Ion Polymer Battery - 3.7V 2500 mAh
1	S2B-PH-K-S	455-1719-ND		0.1	https://www.digi		Connector Header Through Hole, Right Angle 2 position 0.079" (2.00mm)
1	USB Type-C Receptacle (Power-Only)	2073-USB4105-GF-ACT-ND		0.78	https://www.digi		CONN RCP USB2.0 TYP C 24P SMD RA
2	RC0603JR-0722KL	311-22KGRCT-ND		0.2	https://www.digi		22 kOhms ±5% 0.1W, 1/10W Chip Resistor 0603 (1608 Metric) Moisture Resis
2	CL21A226KQQNNNE	1276-2907-1-ND		0.4	https://www.digi		22 µF ±10% 6.3V Ceramic Capacitor X5R 0805 (2012 Metric)
2	CL21A106KAYNNNE	1276-2891-1-ND		0.3	https://www.digi		10 µF ±10% 25V Ceramic Capacitor X5R 0805 (2012 Metric)
1	1239AS-H-1R0M=P2	490-10564-1-ND		0.25	https://www.digi		1 µH Shielded Drum Core, Wirewound Inductor 3 A 59mOhm Max 1008 (252(
1	7440700047	732-1877-1-ND		1.92	https://www.digi		4.7 µH Shielded Drum Core, Wirewound Inductor 3 A 45mOhm Max Nonstan
1	OLED Display - 326	1528-1220-ND		17.5	https://www.digi		Graphic LCD Display Module White OLED - Passive Matrix I2C, SPI 0.96" (24.3
2	RC0603JR-074K7L	311-4.7KGRCT-ND		0.2	https://www.digi		RES 4.7K OHM 5% 1/10W 0603
6	Test Points - 5016	36-5016CT-ND		2.7	https://www.digi		PC Test Point, Compact Phosphor Bronze Silver Plating Surface Mount Mount
1	313070001	1597-1414-ND		18.69	https://www.digi		Adafruit Solar Panel 5V 1.2W
1	B2B-PH-K-S	455-1704-ND		0.1	https://www.digi		Connector Header Through Hole 2 position 0.079" (2.00mm)

Main IC: BQ24075RGTR (U2)

Purpose:
Handles solar-powered Li-Po battery charging and protection.

How it works:

- Solar Panel (P1) supplies power to the input.
- Schottky diode (D1, SS14) prevents reverse current flow from the battery back to the panel.
- Fuse (F1) provides protection against short circuits or overcurrent.
- Capacitor (C1, 10 µF) smooths input voltage.
- Resistors (R1–R4, R13–R16) set charge current limits, enable pins, and status outputs.
- Li-Po Battery (BT) connects to the BAT pin; it's charged to about 3.7 V.
- Labels and power nets mark connected signals and power.

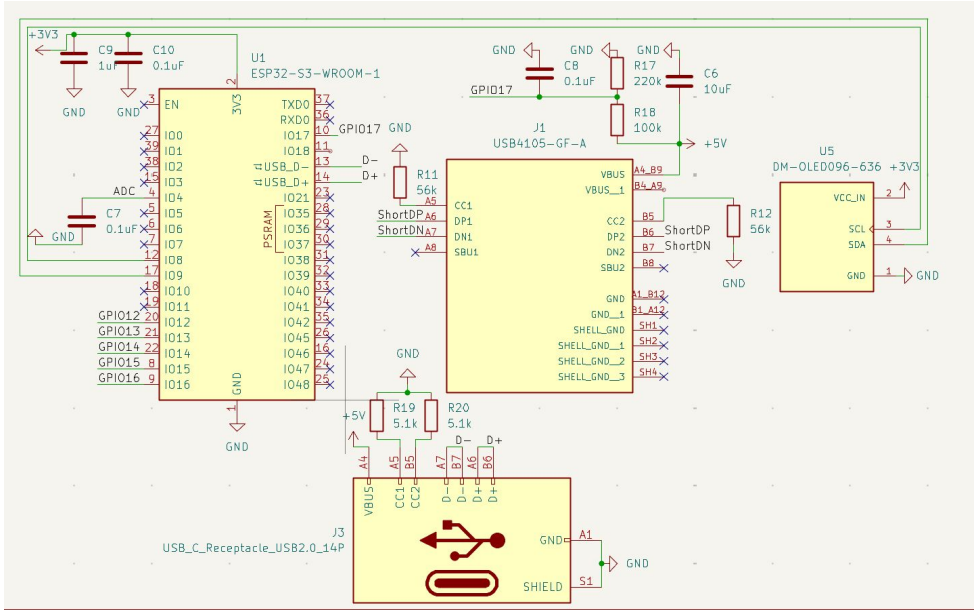


Microcontroller: ESP32-S3-WROOM-1 (U1)

Purpose:
Provides computation, control, and user feedback.

How it works:

- ESP32 runs firmware that:
  - Reads battery data via ADC.
  - Drives the OLED display to show charging status and battery percentage.
  - Can control output enable pins for converters.
- USB4105 (J1) provides USB connectivity to external device.
- USBC (J3) connects to laptop for programming esp32
- Resistors are pull-ups and current limiters for I²C and USB lines.
- Capacitors provide decoupling for stable MCU operation.
- OLED Display (U5) uses I²C (SDA, SCL) to show system status.

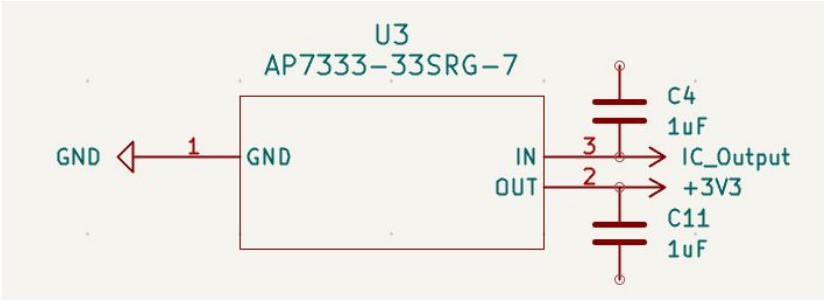


LDO: AP7333-33SRG-7 (U3)

Purpose:
Steps battery voltage (~3.7 V) down to 3.3 V for powering the ESP32 and other logic.

How it works:

- As a linear regulator, it maintains a stable 3.3V output with minimal ripple and low dropout voltage as long as input stays above ~3.5 V.
- Capacitors (C4 and C11) ensure stable operations and filter out voltage noise.
- Includes built in current limiting(~300mA) and thermal shutdown protection for reliability.



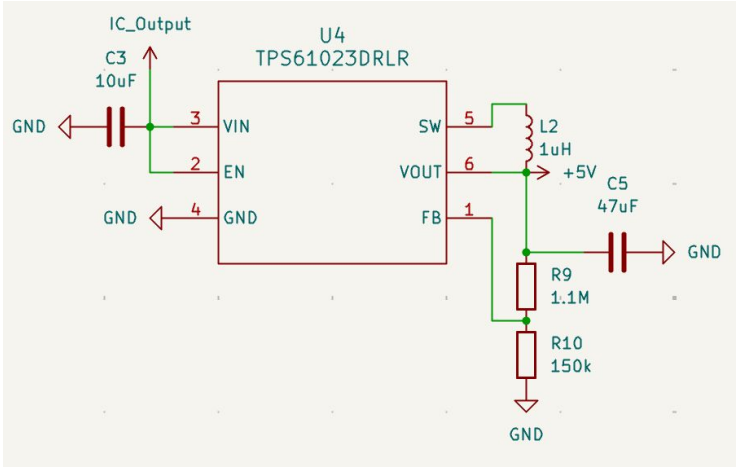
Boost Converter: TPS61023DRLR (U4)

Purpose:

Steps the battery voltage (~3.7 V) up to 5 V to power the USB-C output port and allow charging of external devices.

How it works:

- Acts as a boost converter, using an inductor (L2) and internal switching transistors to store and release energy at a higher voltage.
- When the internal switch turns on, current flows through the inductor and stores energy; when it turns off, that energy is released through a high side MOSFET to raise the output voltage.
- Feedback resistors (R9, R10) tell the IC to maintain a steady 5 V output.
- Capacitors (C3, C5) smooth the voltage and reduce noise, ensuring a clean 5 V rail for USB power.



Layout

As of now, we are trying to finalize our schematic before we can begin layout. Once the layouts are complete, we will move on to component layout and routing within KiCad.

Additionally, we went ahead to estimate the trace widths in mils for power lines for us to prepare for layout. We estimating that High-current paths such as the 5 V USB output could use 60 mil, while battery and solar input wires can use 30 or 40 mils, and the 3.3 V logic rail will use 12–15 mil traces to power the ESP32 and OLED, all other wires remain between 6–8 mil for clean routing.